

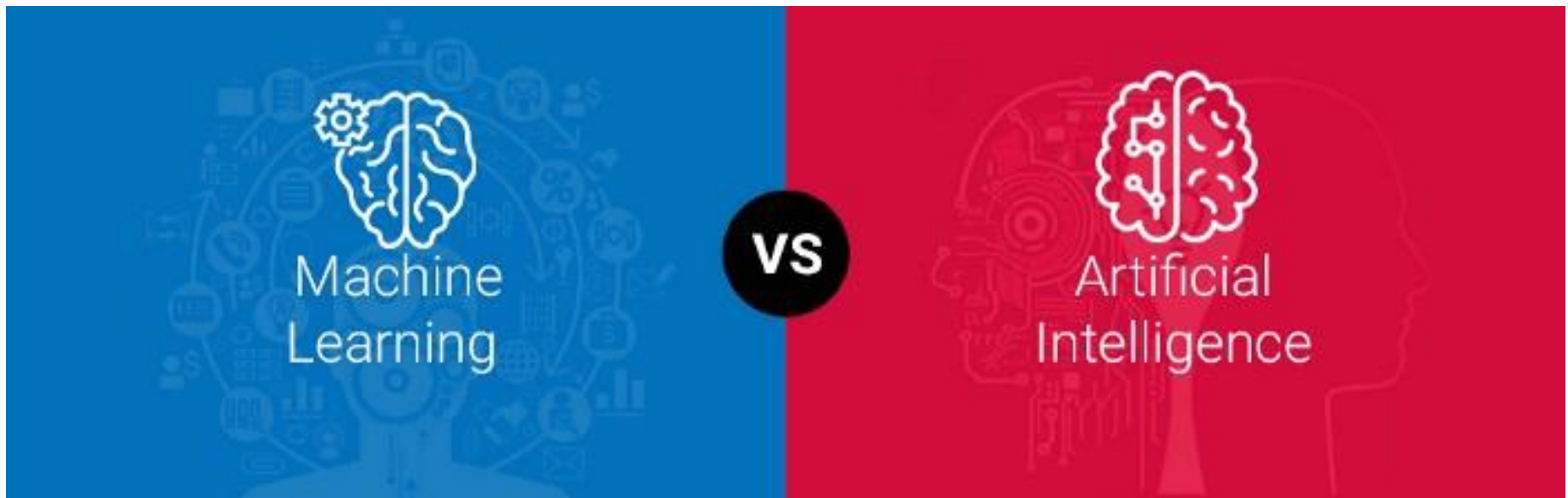
Materi Kuliah (pertemuan ke-5) :

Kecerdasan buatan dan pembelajaran mesin FSM

Dr. Fresy Nugroho, S.T., M.T.

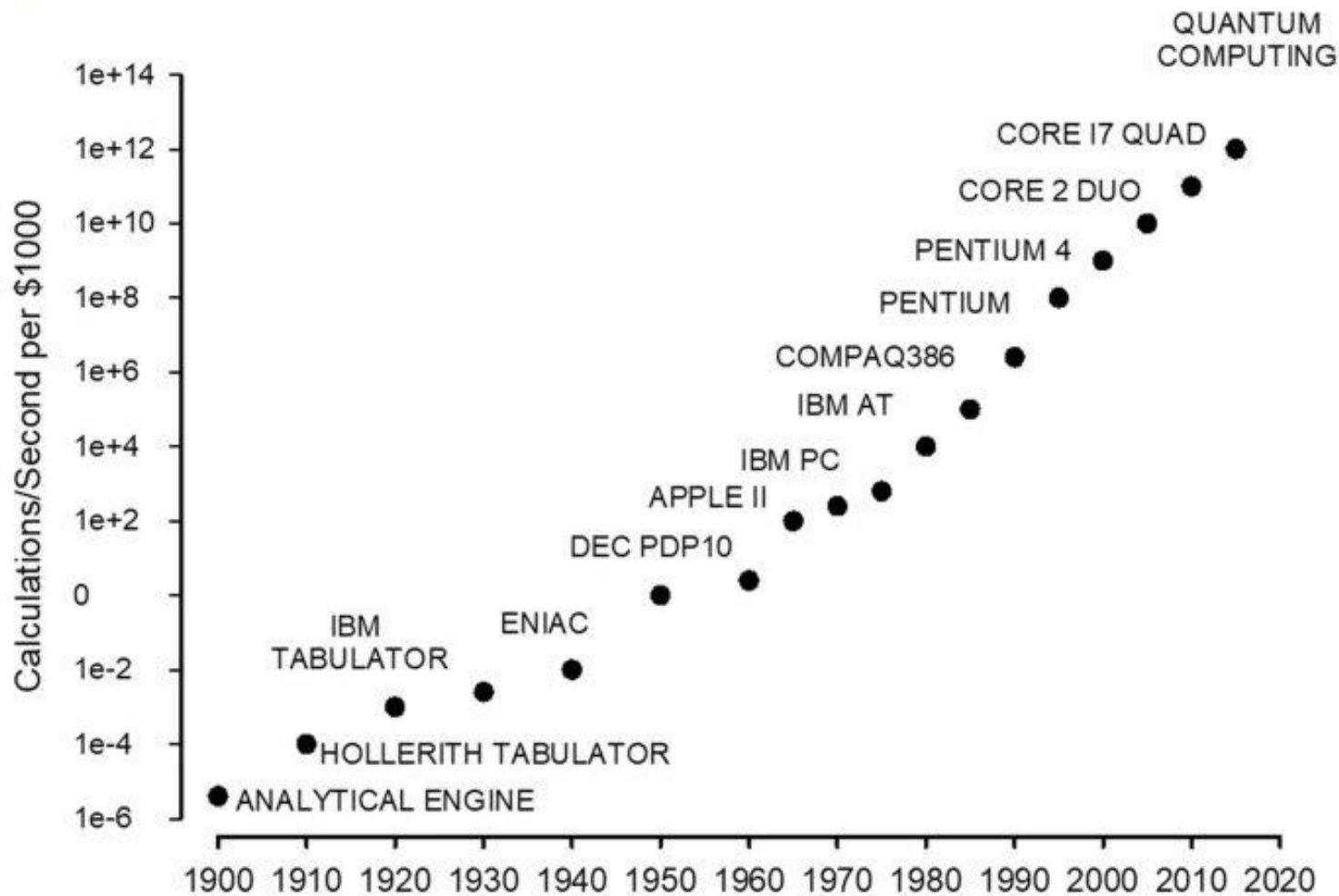
Jurusan Teknik Informatika Fakultas Sains dan Teknologi
Universitas Islam Negeri Maulana Malik Ibrahim Malang
Agustus 2022

- Secara umum, metode yg digunakan dalam pembuatan game adalah :
 - Kecerdasan Buatan (*Artificial Intelligence*) atau
 - Pembelajaran Mesin (*Machine Learning*)



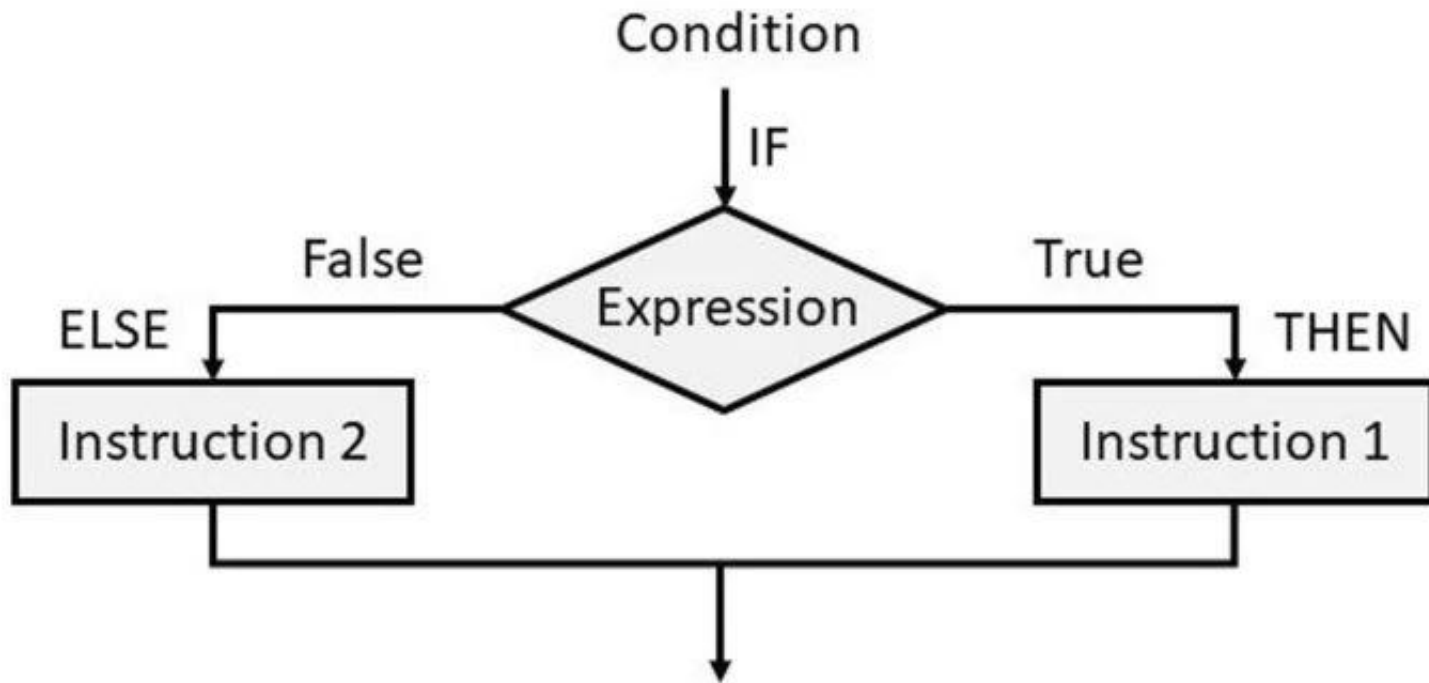
Perkembangan komputer

Fig. 1

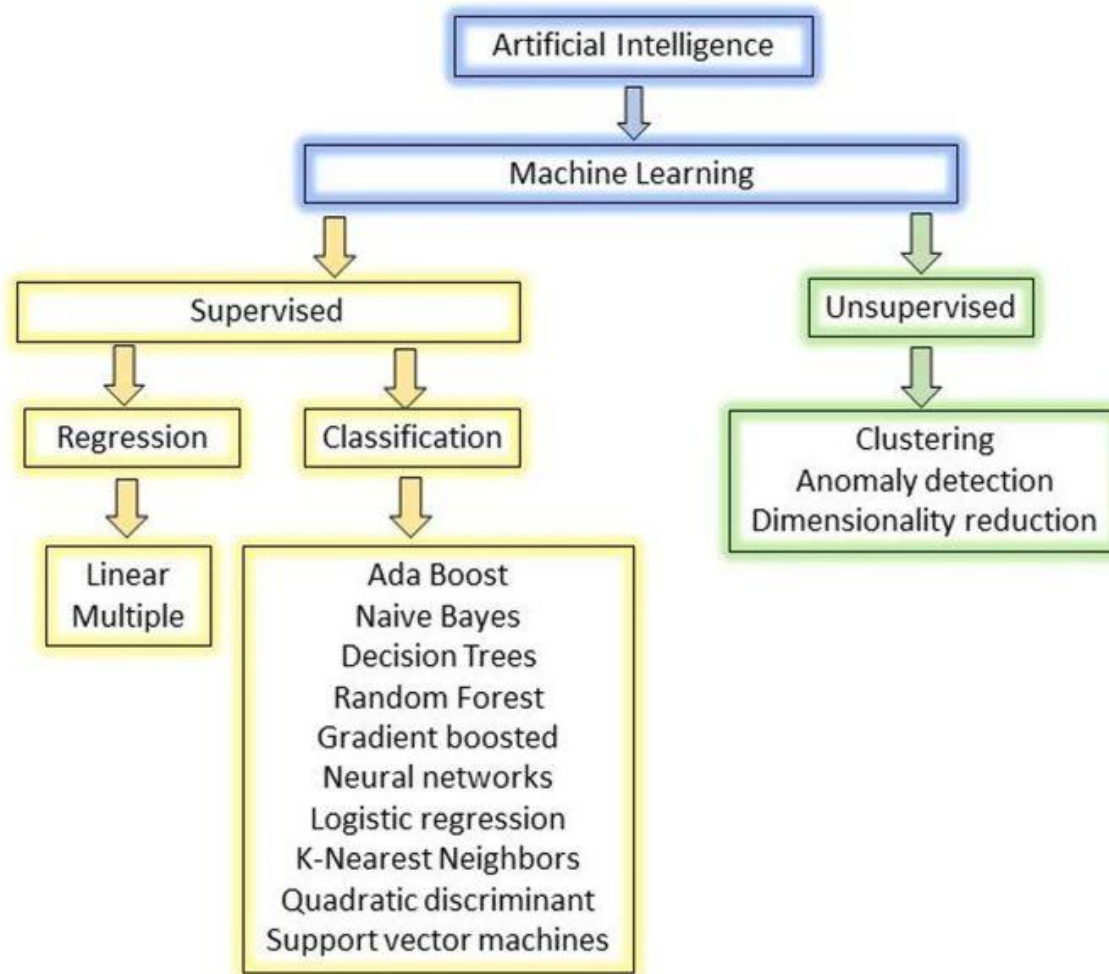


Pembelajaran mesin

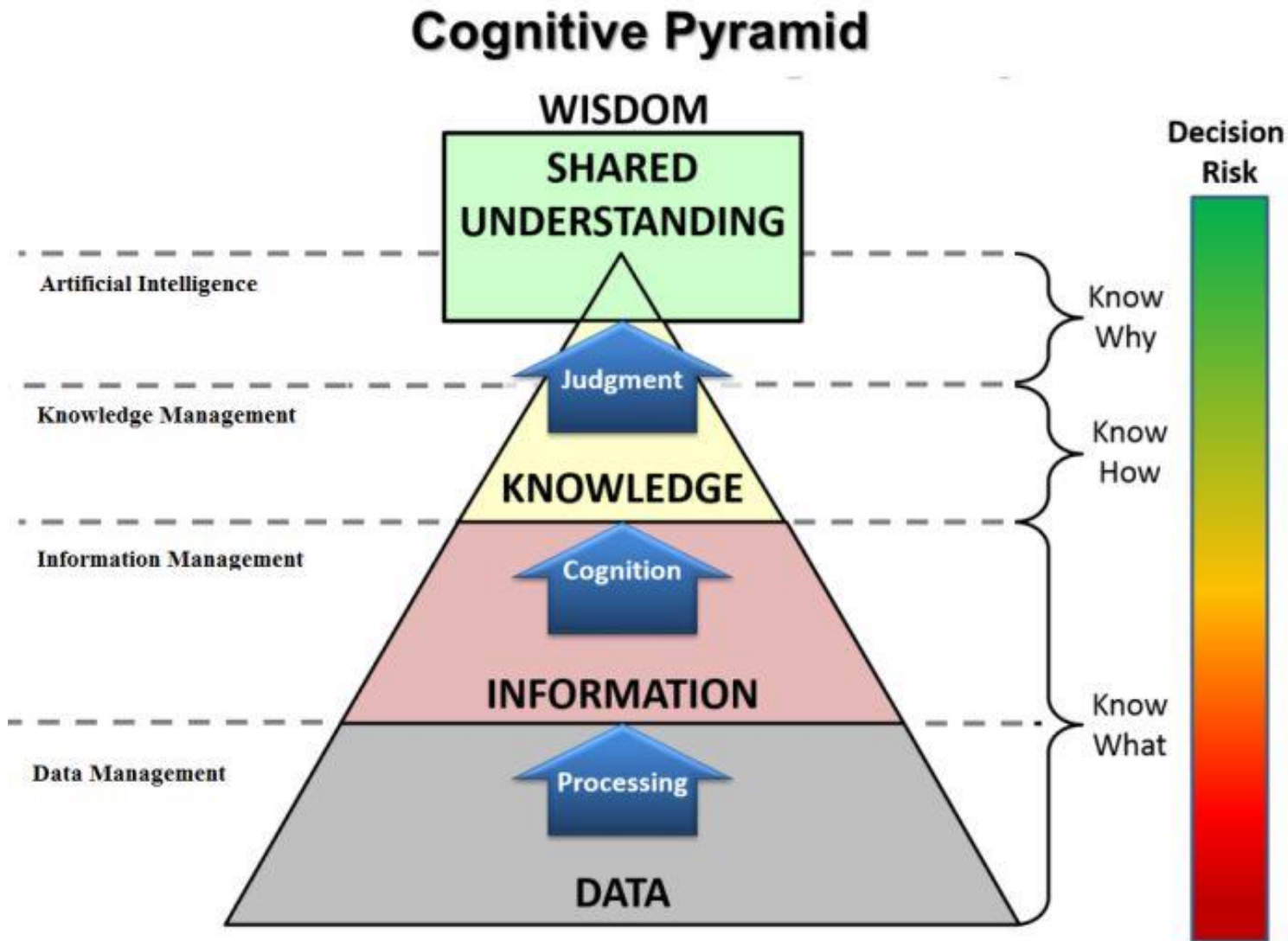
Fig. 2



Kecerdasan Buatan



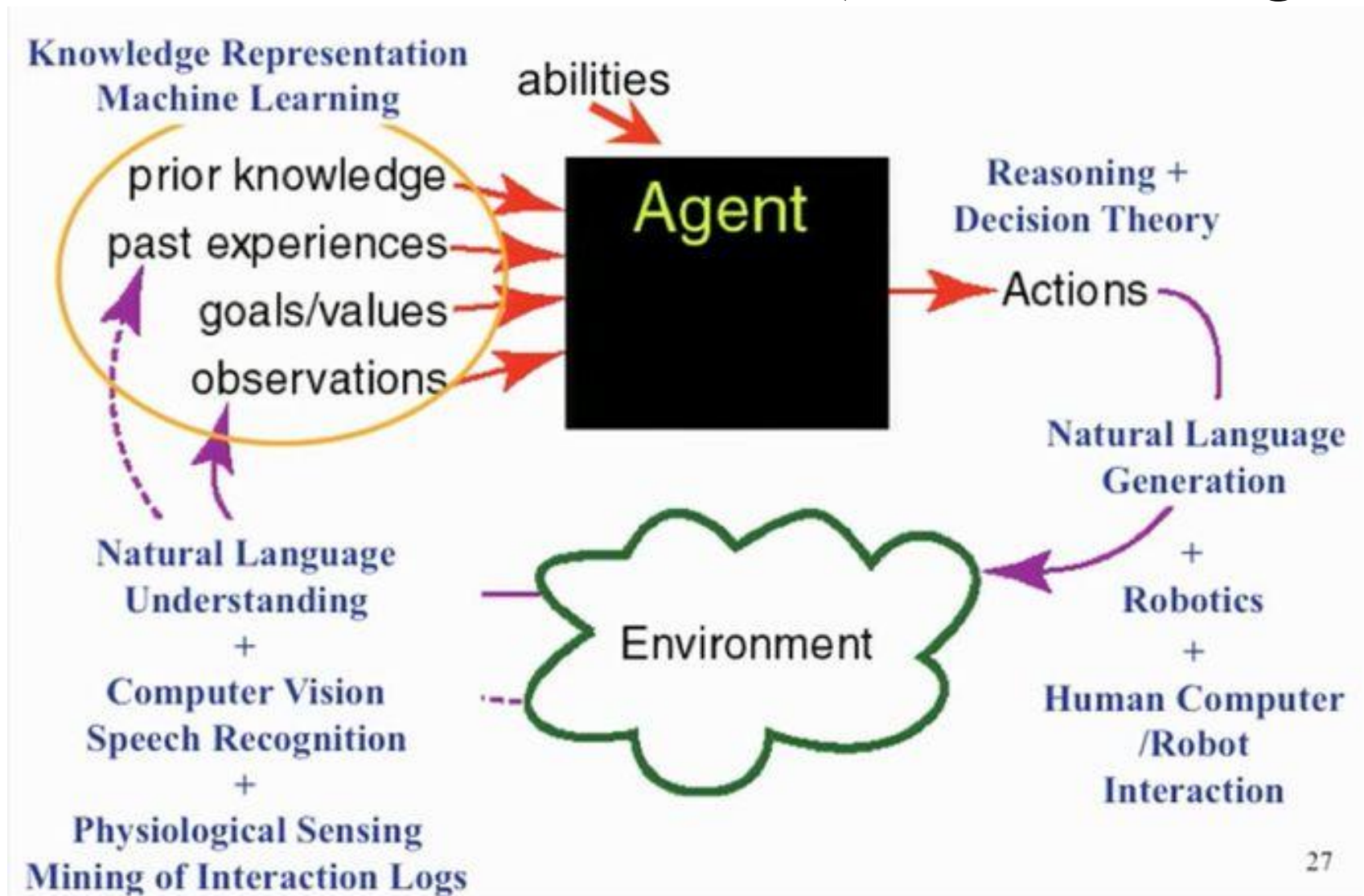
Kecerdasan Buatan (Artificial Intelligence)



Liew, A. (2007). [Understanding Data, Information, Knowledge And Their Inter-Relationships". Journal of Knowledge Management Practice](#). 8 (2). January 2009.

Knight, W. (2017). 5 Big Predictions for Artificial Intelligence in 2017. Retrieved from: <https://www.technologyreview.com/s/603216/5-big-predictions-for-artificial-intelligence-in-2017/>.

Kecerdasan Buatan (Artificial Intelligence)

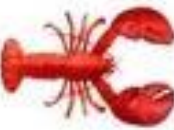


27

Liew, A. (2007). [Understanding Data, Information, Knowledge And Their Inter-Relationships](#)". [Journal of Knowledge Management Practice](#). **8** (2). January 2009.

Knight, W. (2017). 5 Big Predictions for Artificial Intelligence in 2017. Retrieved from: <https://www.technologyreview.com/s/603216/5-big-predictions-for-artificial-intelligence-in-2017/>.

Contoh Klasifikasi

	Feature 1	Feature 2	Feature 3	Feature 4	...	Feature n	Classes
Instance 1	Land	Legs	Mammal	Barks	...	n1	 dog
Instance 2	Land	Legs	Mammal	Purrs	...	n2	 cat
Instance 3	Ocean	Flippers	Mammal	Sings	...	n3	 whale
Instance 4	Ocean	Flippers	Fish	Silent	...	n4	 shark
Instance 5	Ocean	Claws	Crustacean	Silent	...	n5	 lobster

Gutierrez, G. Artificial Intelligence in the Intensive Care Unit. *Crit Care* **24**, 101 (2020).

<https://doi.org/10.1186/s13054-020-2785-y>

Proses membuat pembelajaran mesin

Define the feature matrix
and class vector



Get more data



Split randomly

80% Train



20% Test



Test the model



Evaluate its accuracy



Done

Train the ML algorithm



Use another
algorithm



Teori *Finite State Machine* (FSM)

Symbols - a, b, c, d, 0, 1, 2, 3, ..
Alphabet - Σ (sigma) – Kumpulan dari Symbols. Contoh: {0,1}, {a, b, c, d}
String - Sequence dari symbols. Contoh: a, b, aa, ab, 0, 1, 00, 11, ..
Language - Set of strings
 contoh: $\Sigma = \{0,1\}$

L1 = set of all strings of length 2

 = { 00, 01, 10, 11 }

L2 = set of all string of length 3

 = { 000, 001, 010, 011, 100, 101, 110, 111 }

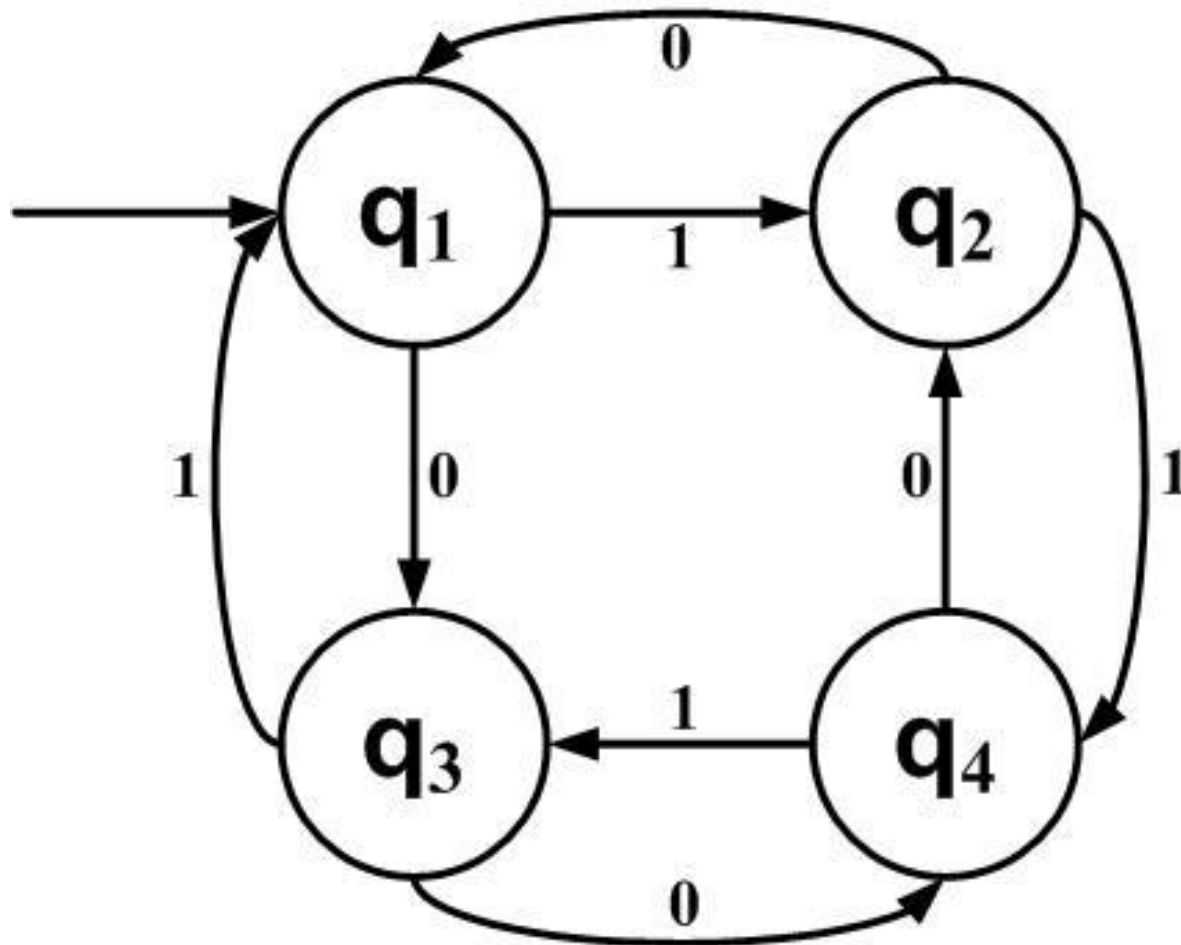
L3 = set of all string begin with 0

 = { 0, 00, 01, 000, 001, 010, 011, 0000, 0001, }

L 1 & L2 = Finite sets

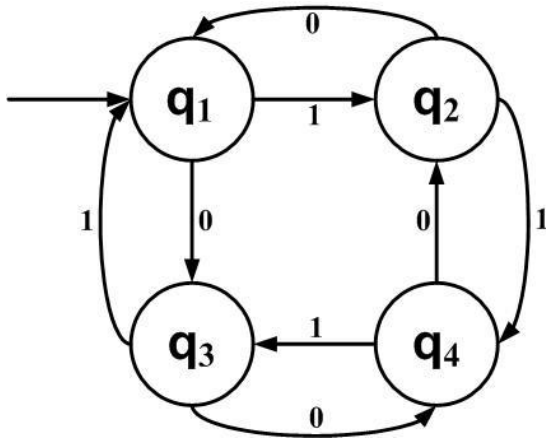
L3 = Infinite sets

Teori *Finite State Machine* (FSM)



V. Vaid, “Comparison of different attributes in modeling a FSM based vending machine in 2 different styles,” no. Ices, pp. 18–21, 2014.

Teori *Finite State Machine* (FSM)



Finite State Automata dapat didefinisikan dengan 5 Tuple yaitu $(Q, \Sigma, \delta, S, F)$

Q = himpunan state = $\{q_1, q_2, q_3, q_4\}$

Σ = himpunan symbol input/masukan = $\{0,1\}$

S = state awal = $\{q_1\}$

F = state akhir = $\{q_2\}$

δ = fungsi transisi $Q \times \Sigma \rightarrow Q$

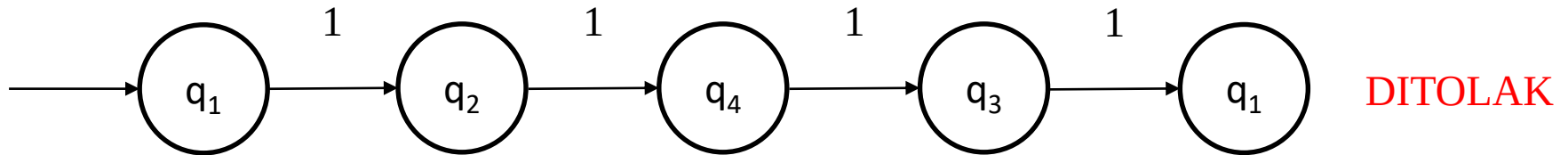
State \ Input	0	1
q_1	q_3	q_2
q_2	q_1	q_4
q_3	q_4	q_1
q_4	q_2	q_3

Teori *Finite State Machine* (FSM)

Sebagai contoh,

Terdapat string 1111, uji apakah string tersebut diterima atau ditolak

State \ Input	0	1
q_1	q_3	q_2
q_2	q_1	q_4
q_3	q_4	q_1
q_4	q_2	q_3

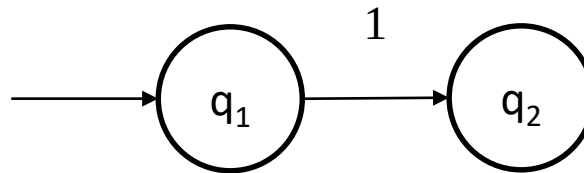


Teori *Finite State Machine* (FSM)

Sebagai contoh,

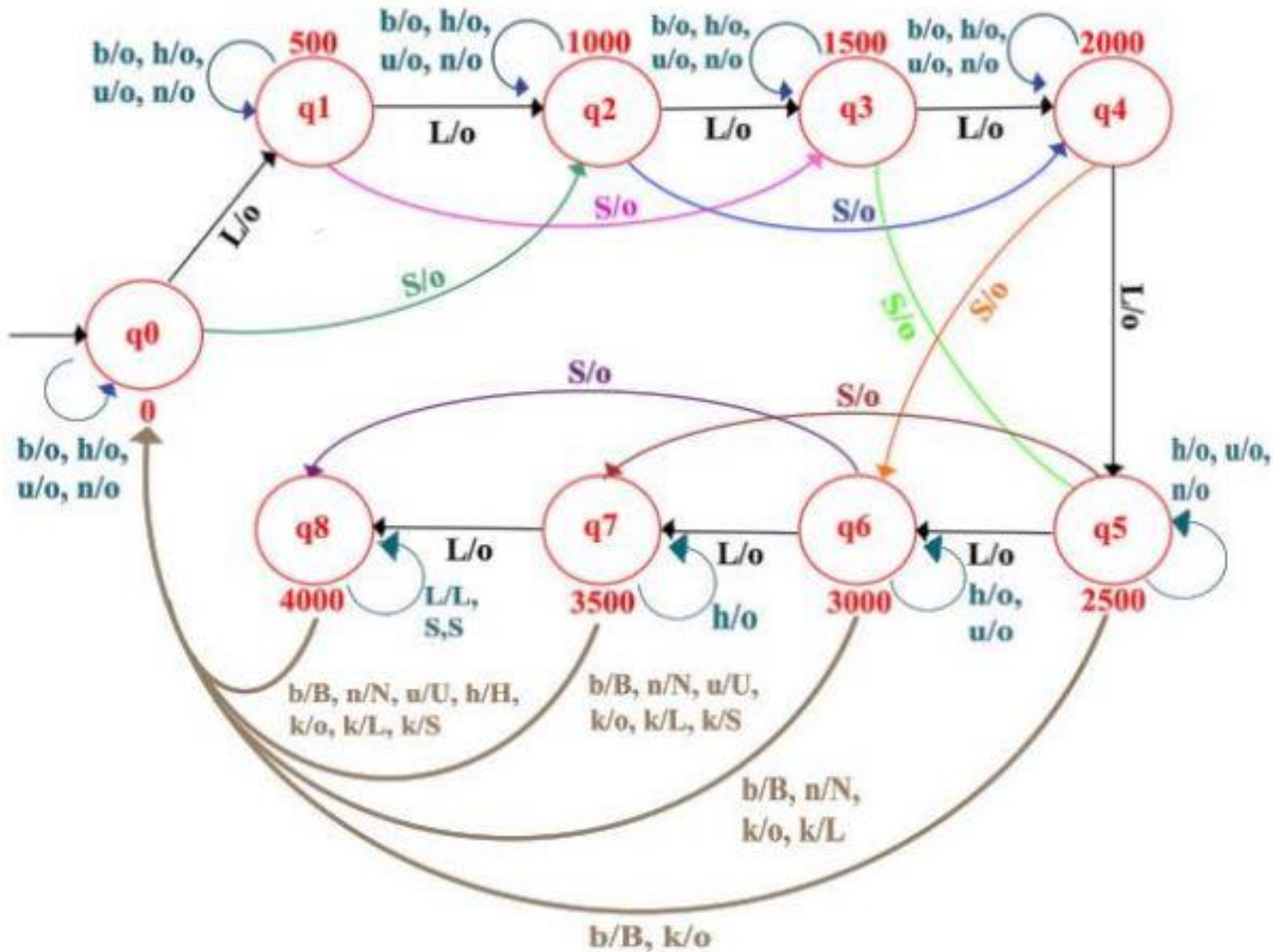
Terdapat string 1, uji apakah string tersebut diterima atau ditolak

State \ Input	0	1
q_1	q_3	q_2
q_2	q_1	q_4
q_3	q_4	q_1
q_4	q_2	q_3

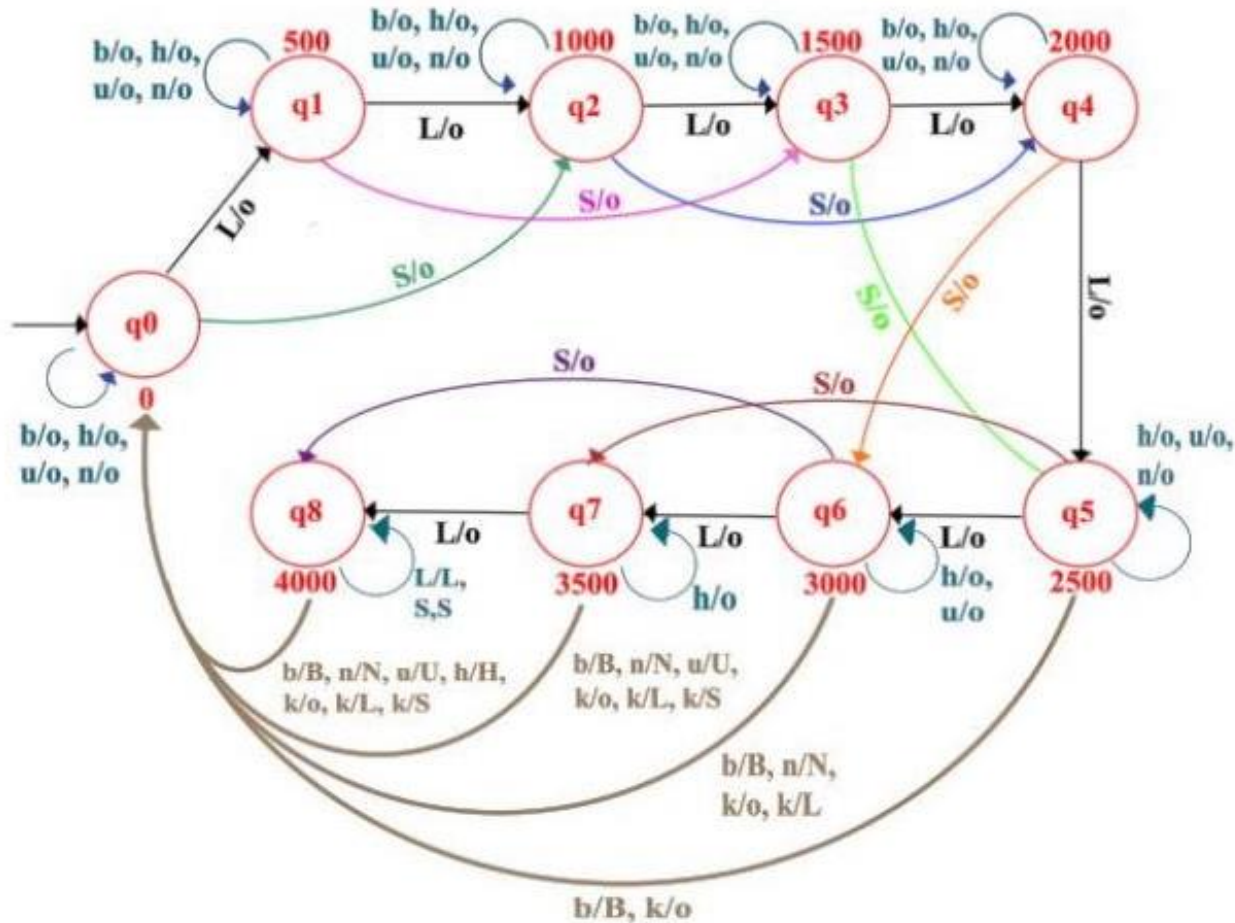


DITERIMA

Finite State Machine (FSM) dunia nyata



Finite State Machine (FSM) dunia nyata



Input:

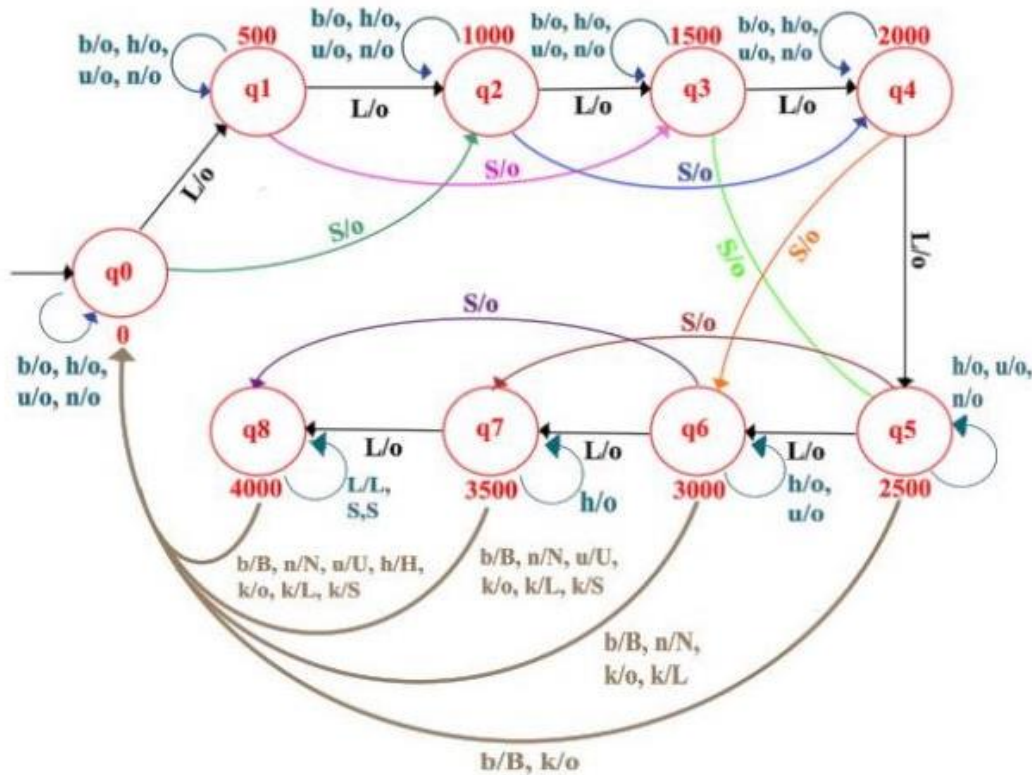
b (pilih Buavita)
n (pilih Nescafe)
u (pilih Ultra Milk)
h (pilih Hilo)
L (logam 500 rupiah)
S (logam 1.000 rupiah)
k (kembalian)

Output:

o (tidak ada)
B (keluar Buavita)
N (keluar Nescafe)
U (keluar Ultra Milk)
H (keluar Hilo)

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$
 $\Sigma = \{b, n, u, h, L, S, k\}$

Finite State Machine (FSM) dunia nyata



Tabel 2: Tabel transisi masukan

δ	b	n	u	h	L	S	k
q_0	q_0	q_0	q_0	q_0	q_1	q_2	q_0
q_1	q_1	q_1	q_1	q_1	q_2	q_3	q_1
q_2	q_2	q_2	q_2	q_2	q_3	q_4	q_2
q_3	q_3	q_3	q_3	q_3	q_4	q_5	q_3
q_4	q_4	q_4	q_4	q_4	q_5	q_6	q_4
q_5	q_0	q_5	q_5	q_5	q_6	q_7	q_0
q_6	q_0	q_0	q_6	q_6	q_7	q_8	q_0
q_7	q_0	q_0	q_0	q_7	q_8	q_8	q_0
q_8	q_0	q_0	q_0	q_0	q_8	q_8	q_0

Finite State Machine (FSM) dunia nyata

Definisi :

S = state awal = $\{q_0\}$

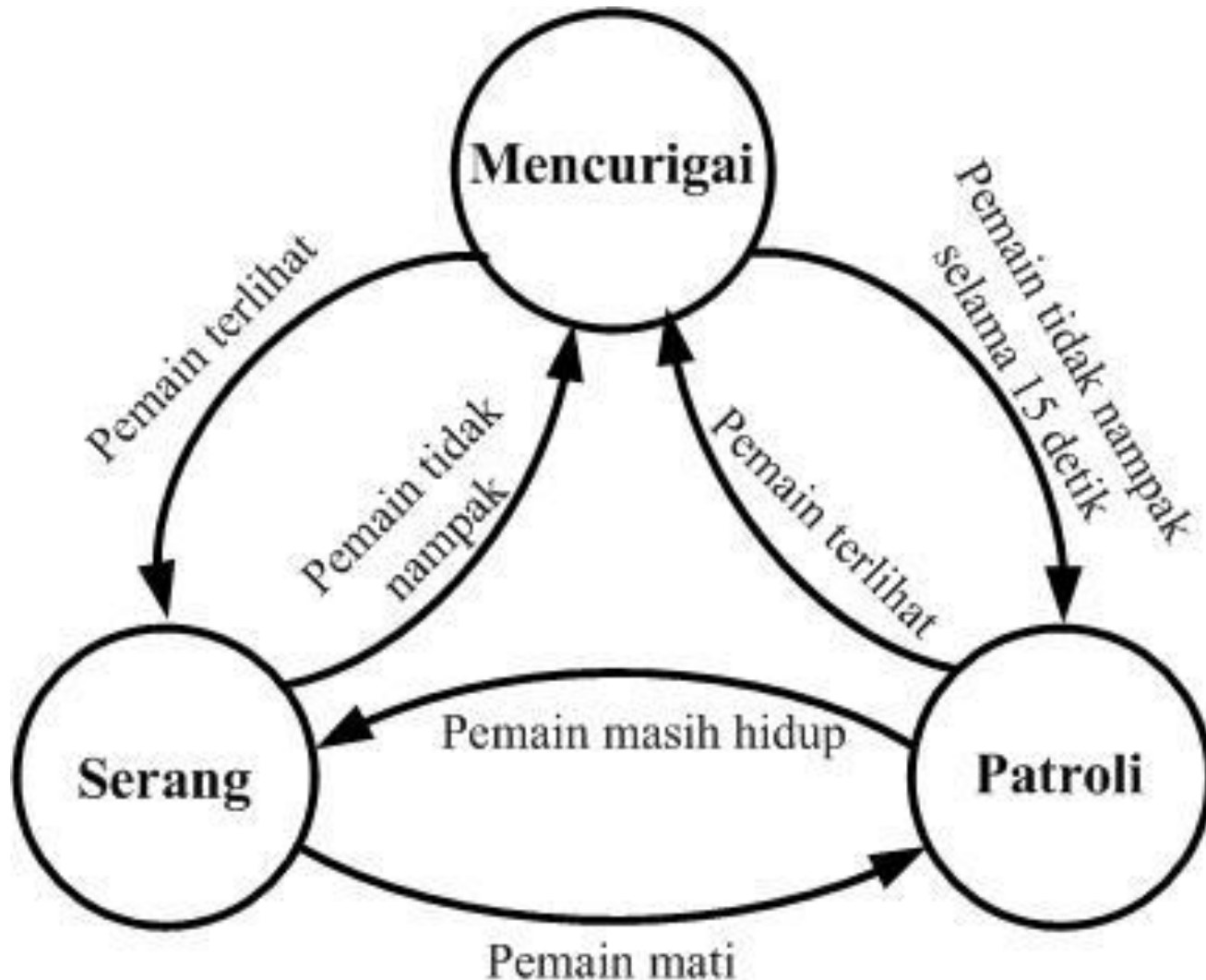
Δ = himpunan symbol keluaran = $\{B, N, U, H, o\}$

λ = fungsi keluaran =

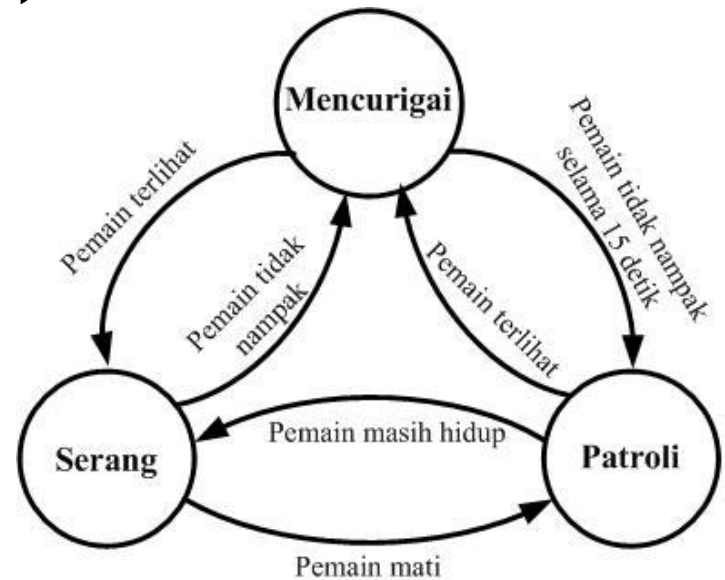
Tabel 3: Tabel fungsi keluaran

δ	b	n	u	h	L	S	k
q_0	o	o	o	o	o	o	o
q_1	o	o	o	o	o	o	o
q_2	o	o	o	o	o	o	o
q_3	o	o	o	o	o	o	o
q_4	o	o	o	o	o	o	o
q_5	B	o	o	o	o	o	o
q_6	B	N	o	o	o	o	o, L
q_7	B	N	U	o	o	o	o, L, S
q_8	B	N	U	H	L	S	o, L, S

Finite State Machine (FSM) dalam game



Finite State Machine (FSM)



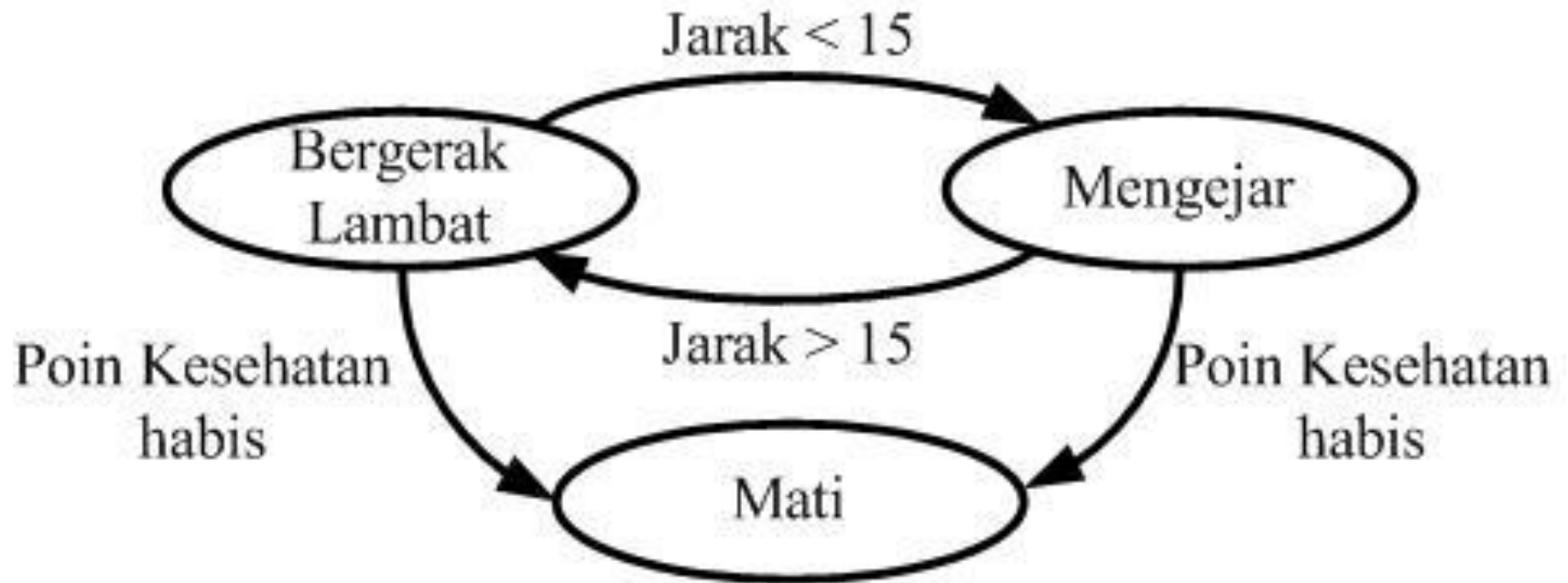
Keuntungan dari FSM :

- a) mudah diimplementasikan dan dipahami.
- b) mudah diperluas, asalkan jumlah state relatif kecil.
- c) Debugging juga menjadi relatif sederhana.

Salah satu kelemahan FSM :

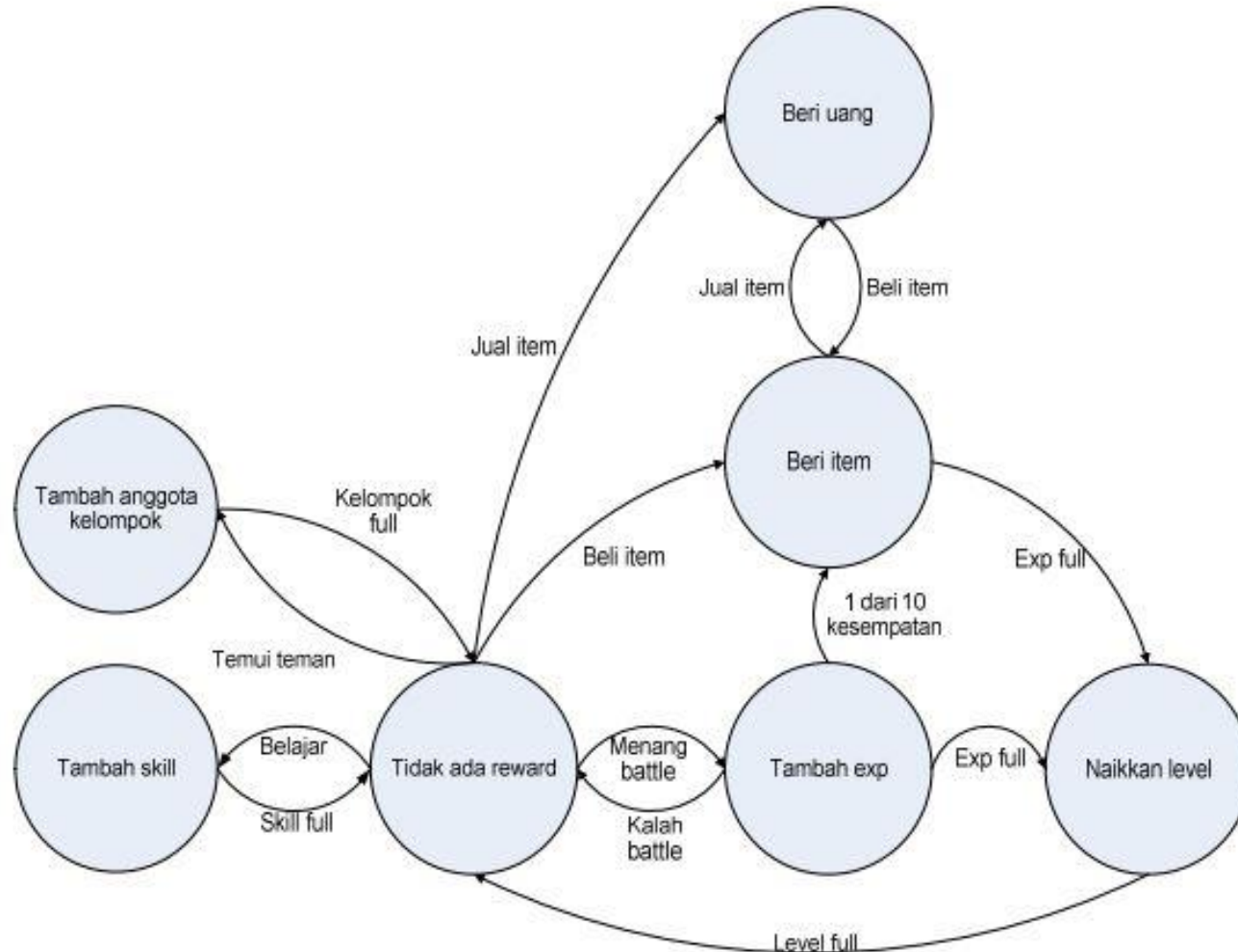
- a) sulit untuk mengelola,
- b) sulit untuk mempertahankan, dan
- c) sulit untuk memperluas seiring dengan bertambahnya jumlah state

Finite State Machine (FSM)



FSM Kategori Musuh A

Finite State Machine (FSM)



Gambar 4 Rancangan FSM untuk Reward Dinamis

Finite State Machine (FSM)

Tabel 1 Matriks Transisi Sistem Reward

To From	Tidak ada reward	Tambah exp	Naikkan level	Tambah skill	Tambah anggota kelompok	Beri uang	Beri item
Tidak ada reward	-	Menang battle	-	Belajar	Temui teman	Jual item	Beli item
Tambah exp	Kalah battle	-	Exp full	-	-	-	1/10
Naikkan level	Level full	-	-	-	-	-	-
Tambah skill	Skill full	-	-	-	-	-	-
Tambah anggota kelompok	Kelompok full	-	-	-	-	-	-
Beri uang	-	-	-	-	-	-	Beli item
Beri item	-	-	Exp full	-	-	Jual item	-

Finite State Machine (FSM)

Tabel 2 Tabel Hasil Perpindahan State Perilaku Reward

Input State	MB	KB	EF	LF	B	SF	T	KF	JI	BI	K
TR	TE	TR	TR	TR	TS	TR	TA	TR	U	I	TR
TE	TE	TR	NL	TE	TE	TE	TE	TE	TE	TE	I
NL	NL	NL	NL	TR	NL	NL	NL	NL	NL	NL	NL
TS	TS	TS	TS	TS	TS	TR	TS	TS	TS	TS	TS
TA	TA	TA	TA	TA	TA	TA	TA	TR	TA	TA	TA
U	U	U	U	U	U	U	U	U	U	I	U
I	I	I	I	I	I	I	I	I	U	I	I

Finite State Machine (FSM)

Keterangan kode dari state pada Tabel 2 di atas adalah sebagai berikut :

- TR : Tidak ada reward
- TE : Tambah Exp
- NL : Naikkan Level
- TS : Tambah skill
- TA : Tambah anggota kelompok
- U : Beri uang

Keterangan kode dari input pada Tabel 2 di atas adalah sebagai berikut :

- MB : Menang battle
- KB : Kalah battle
- EF : Exp full
- LF : Level full
- B : Belajar
- SF : Skill Full
- T : Temui teman
- KF : Kelompok full
- JI : Jual item
- BI : Beli item
- K : 1/10 kesempatan

Hierarchical Finite State Machine (HFSM)

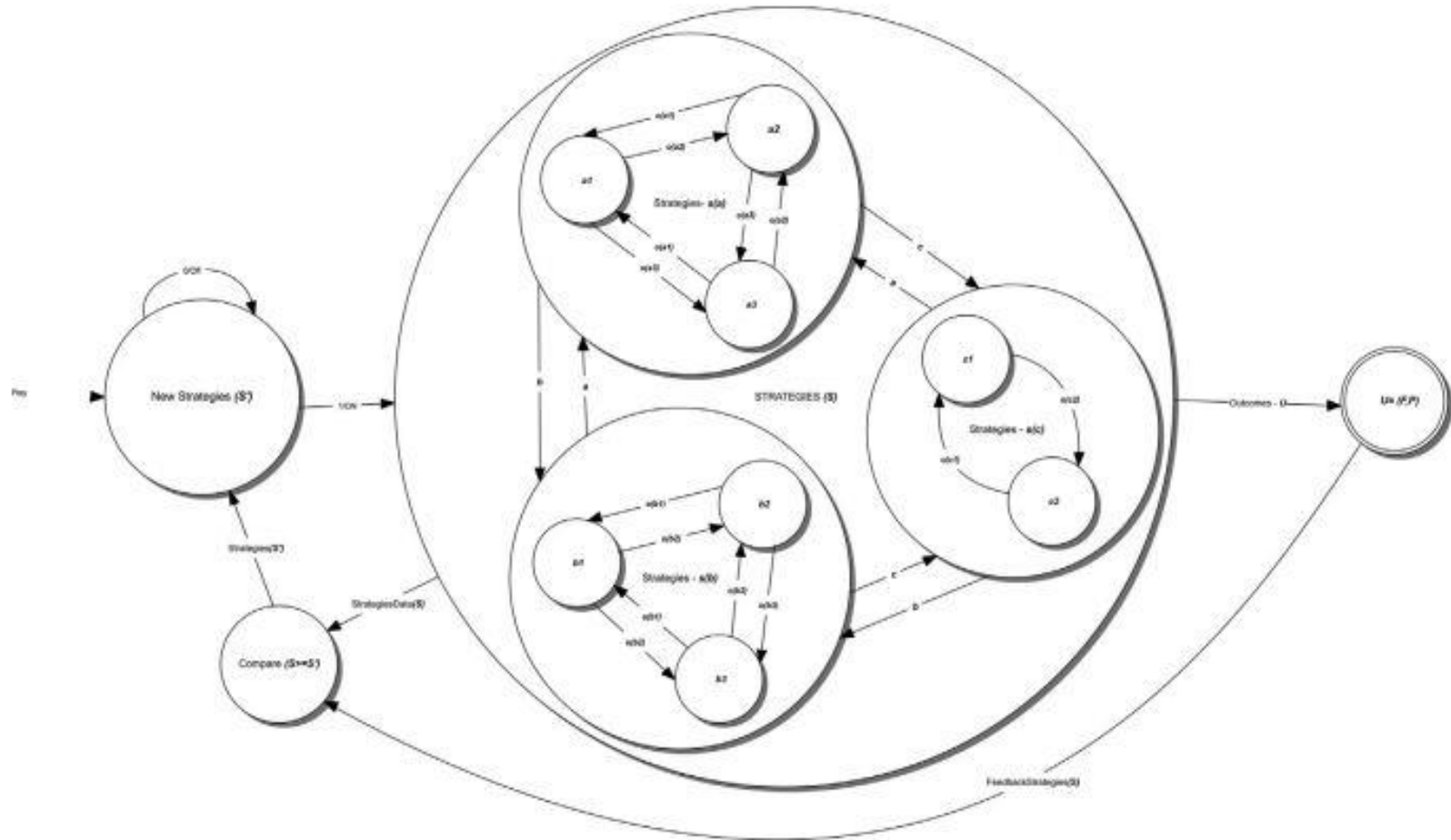


Figure 7. HFSM Serious Game for tillage base on pareto optima.

Hierarchical Finite State Machine (HFSM)

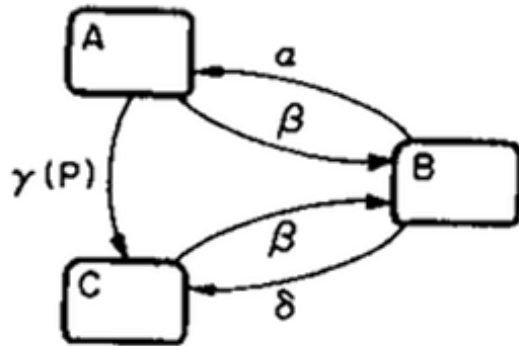


Fig. 1.

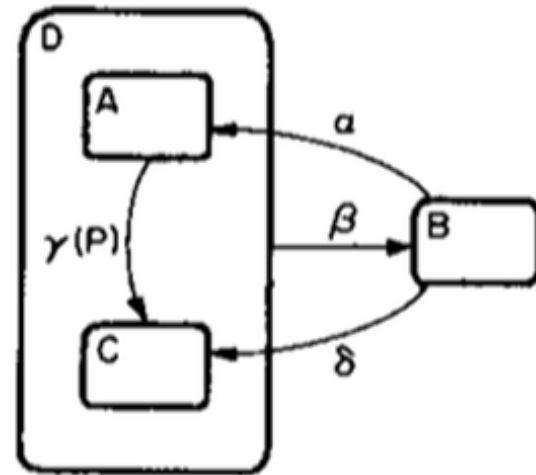
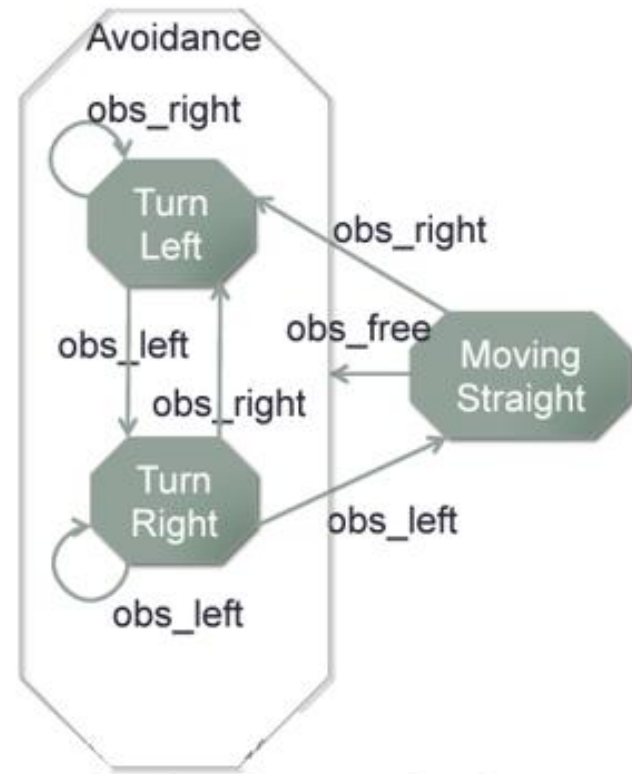
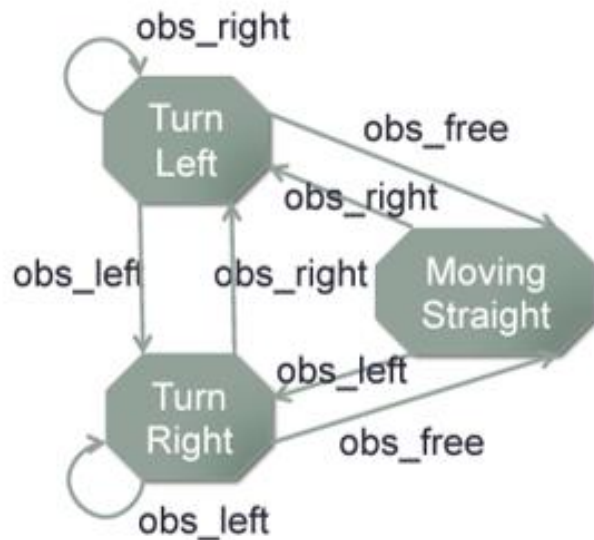


Fig. 2.

Hierarchical Finite State Machine (HFSM)



Implementasi HFSM dunia nyata



Fig. 11. OSU-ACT at DARPA Urban Challenge 2007.

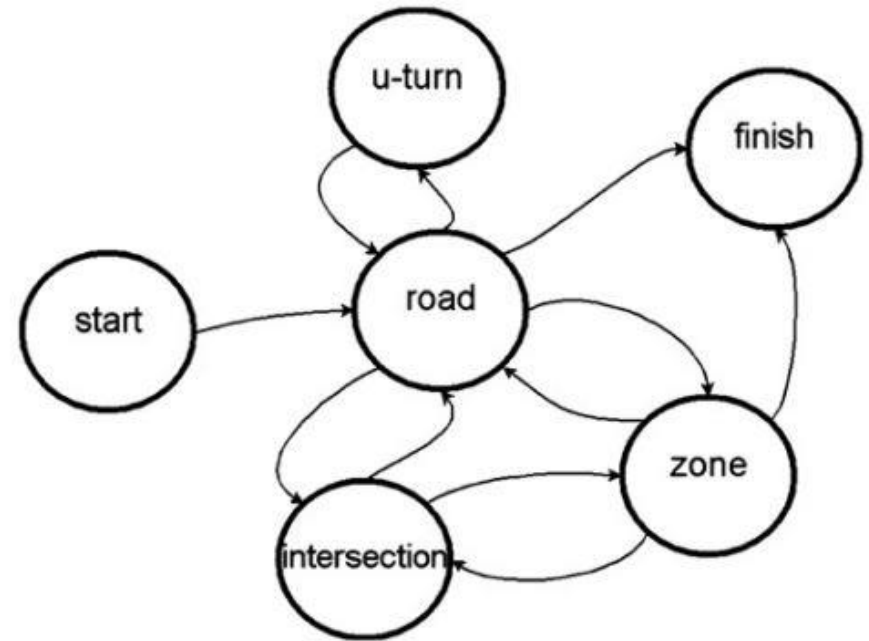
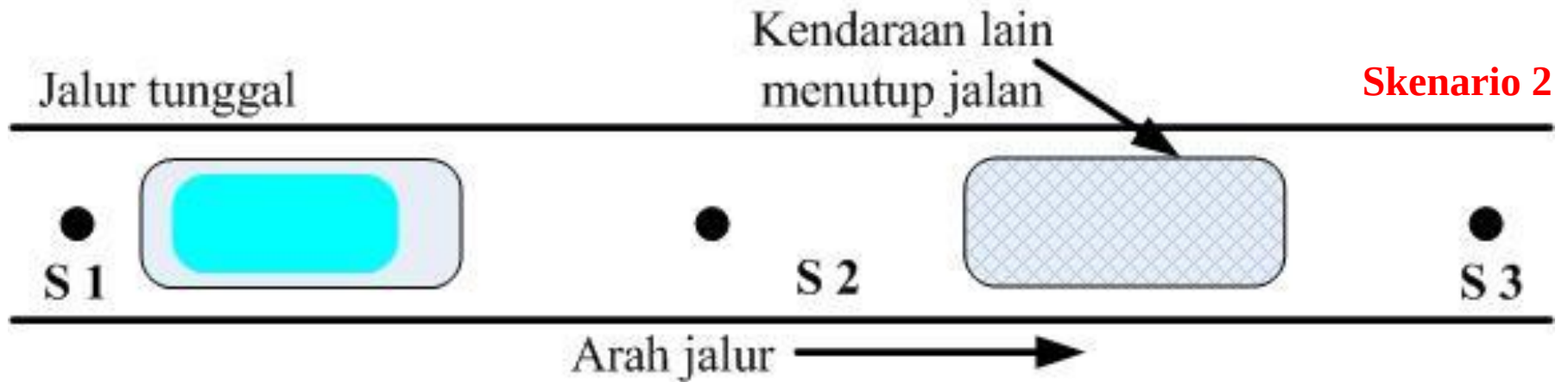
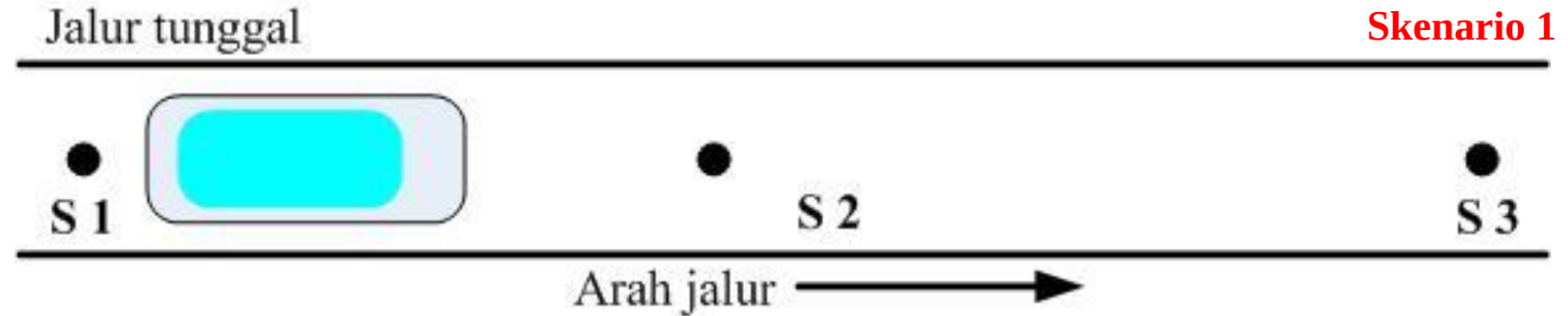


Fig. 12. OSU-ACT meta-state connections.

Implementasi HFSM : Skenario 1 dan 2

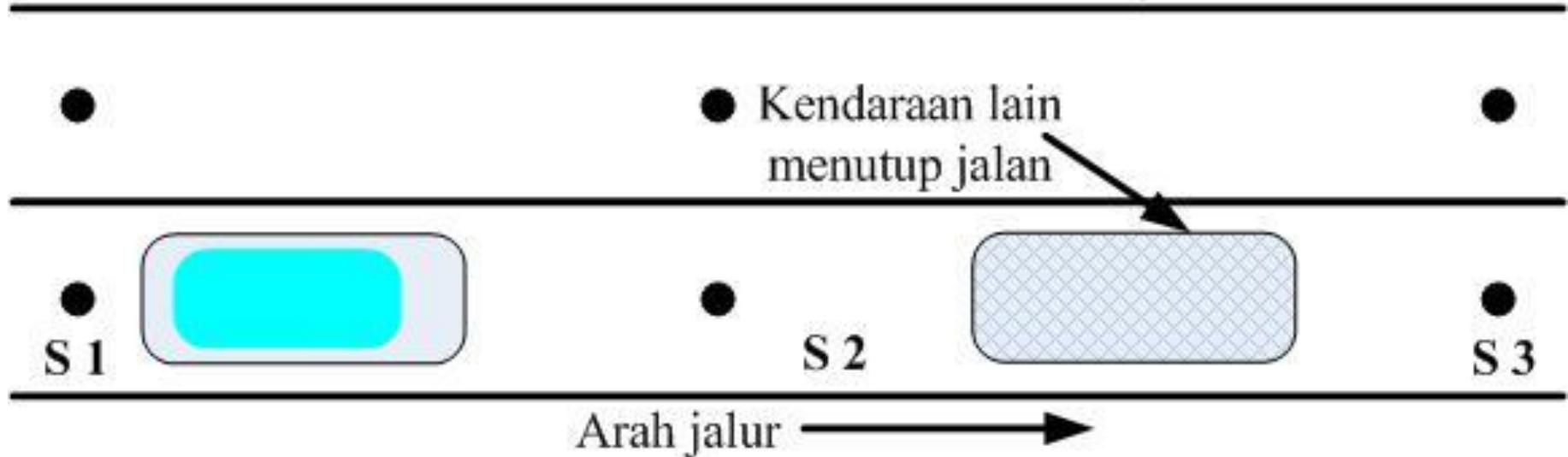


Implementasi HFSM :

Jalur ganda

Arah jalur ←

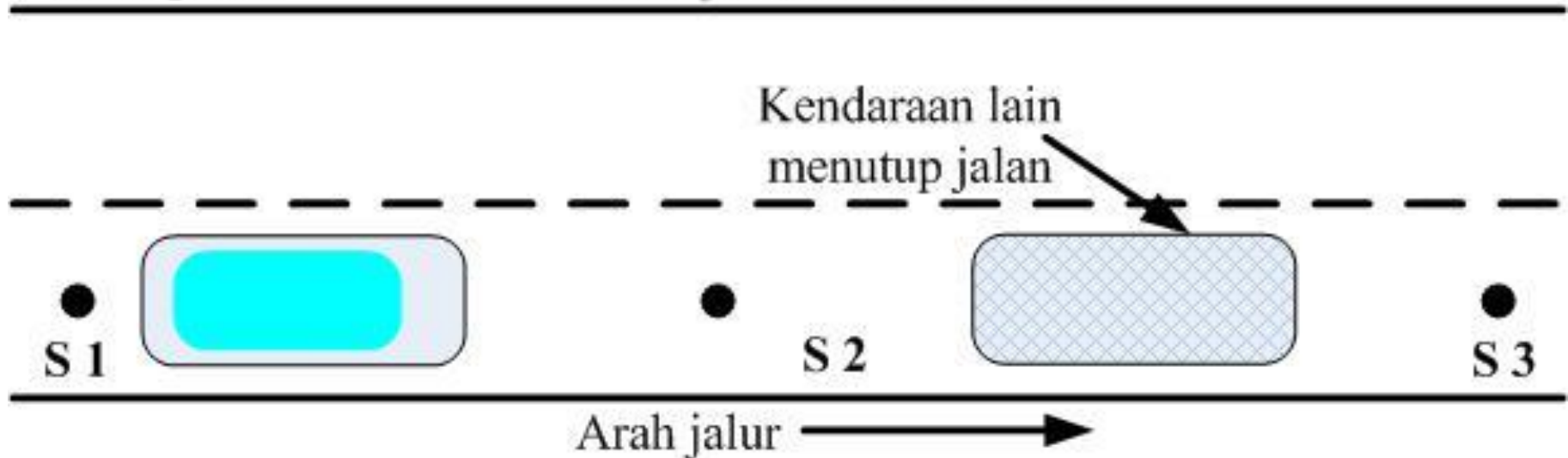
Skenario 3



Jalur ganda

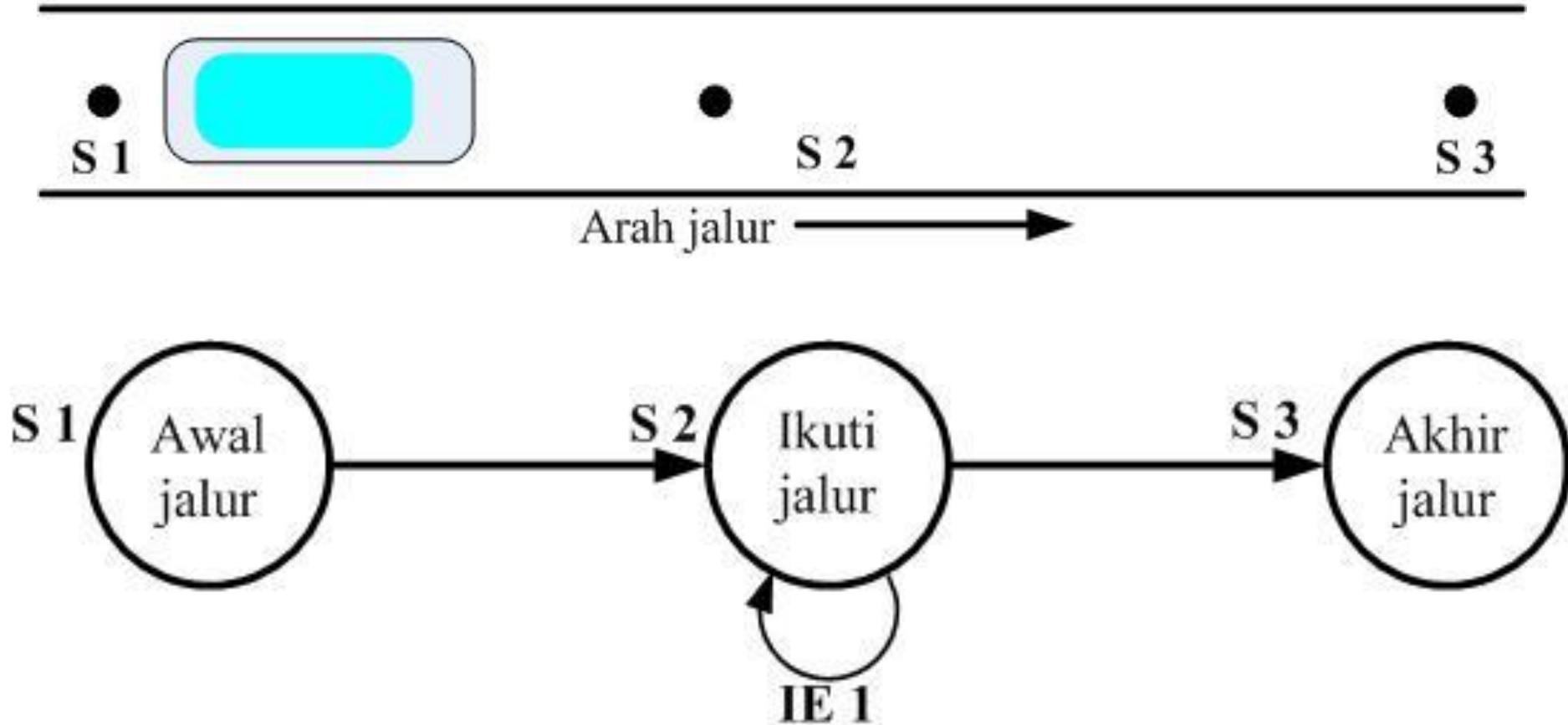
Arah jalur tidak diketahui

Skenario 4

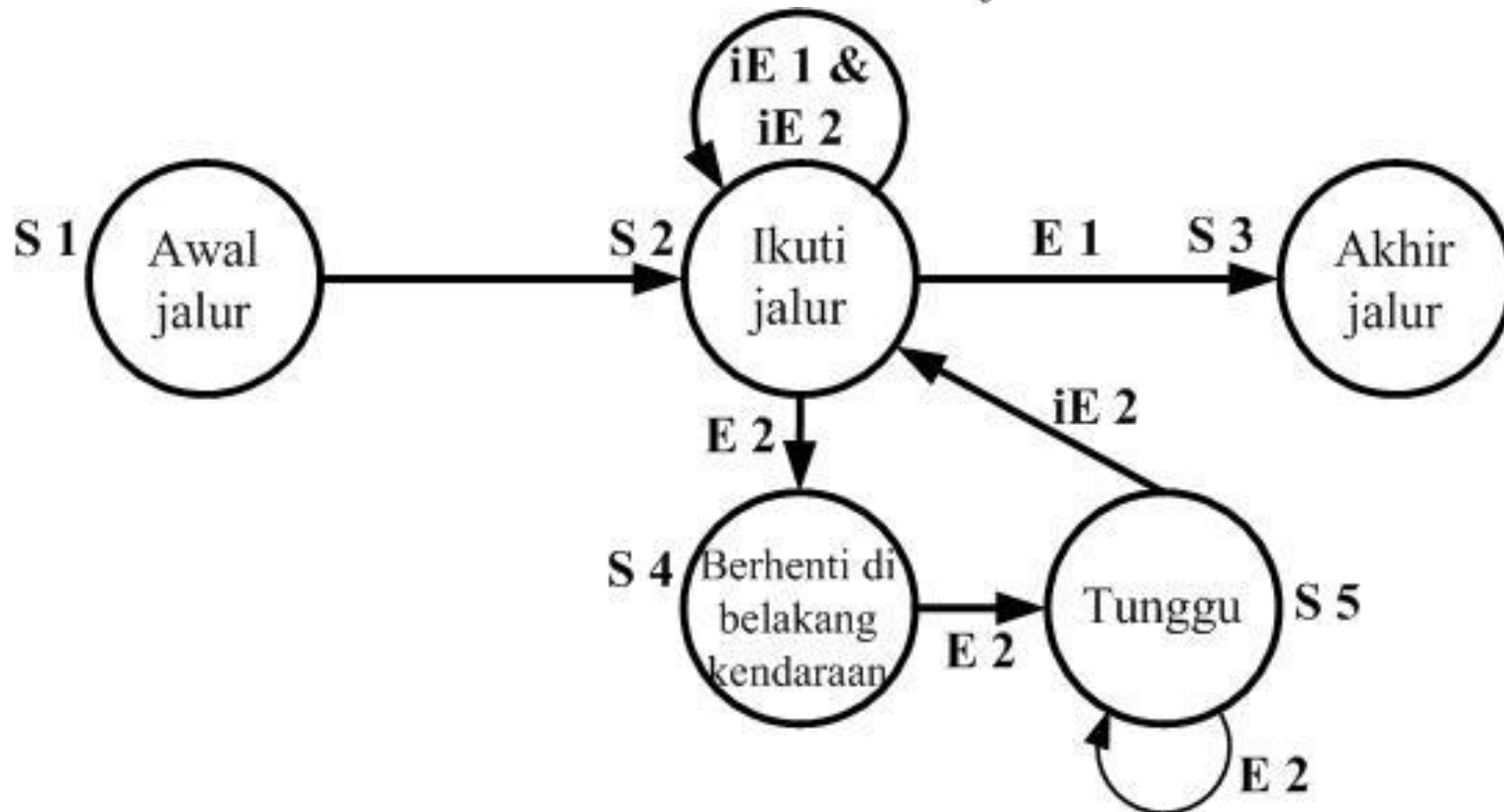
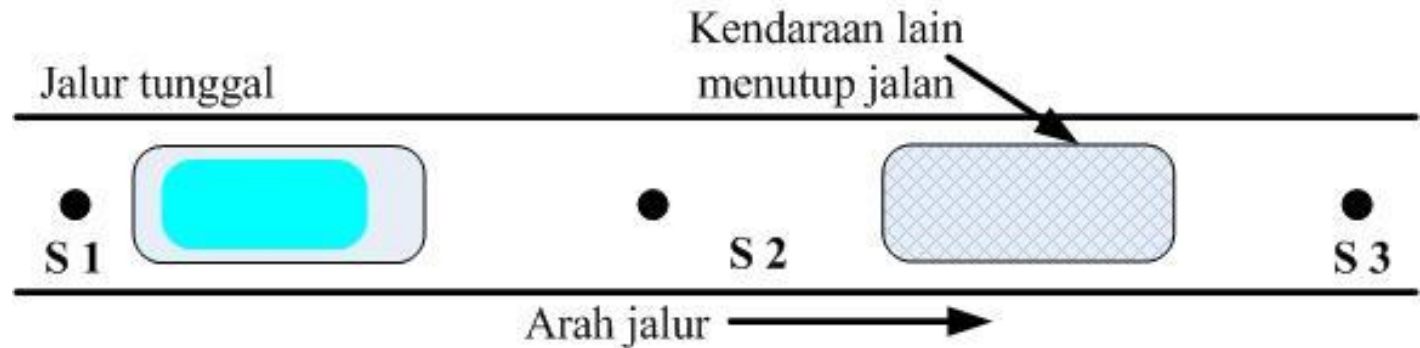


Implementasi HFSM : Skenario 1

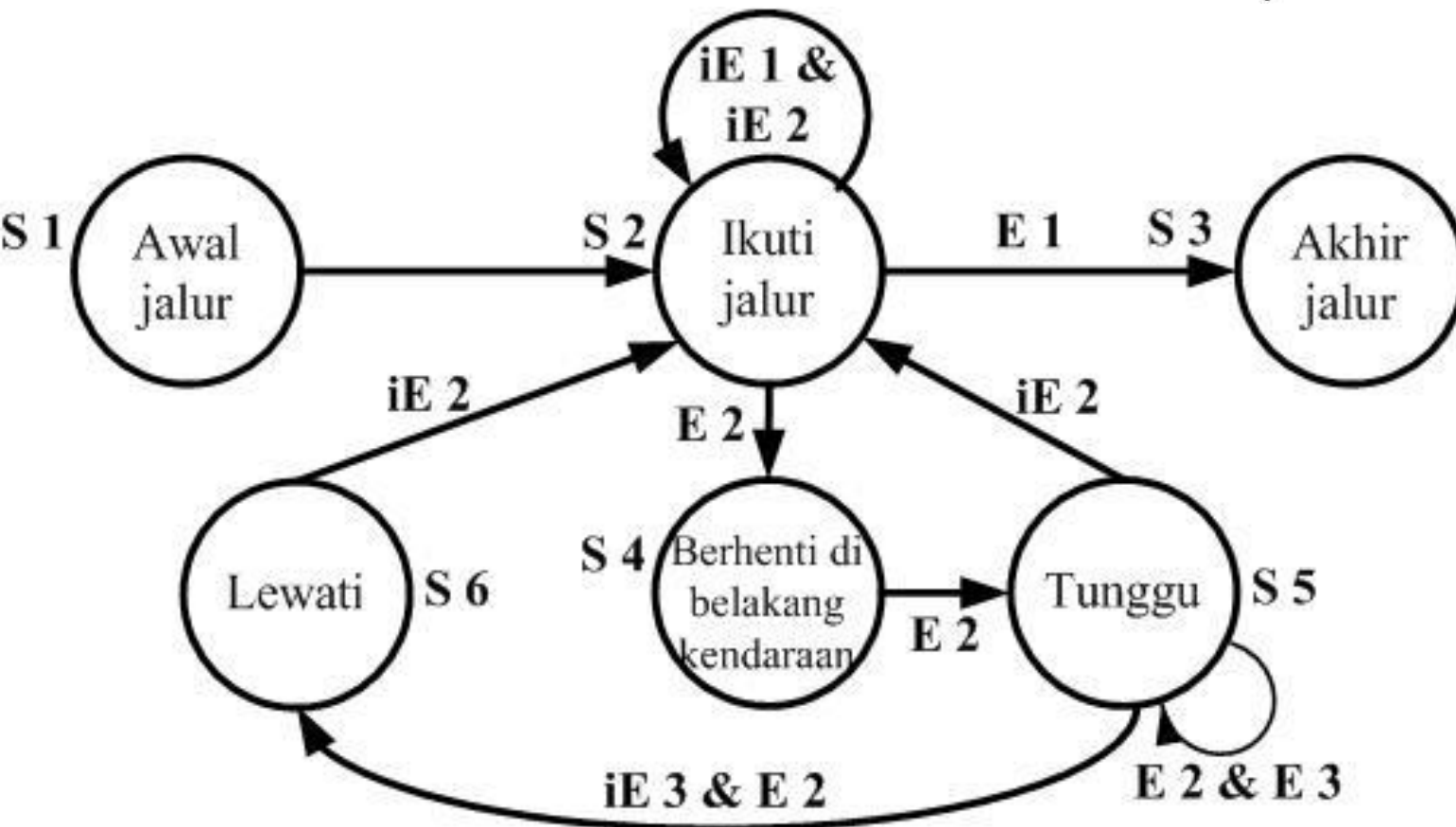
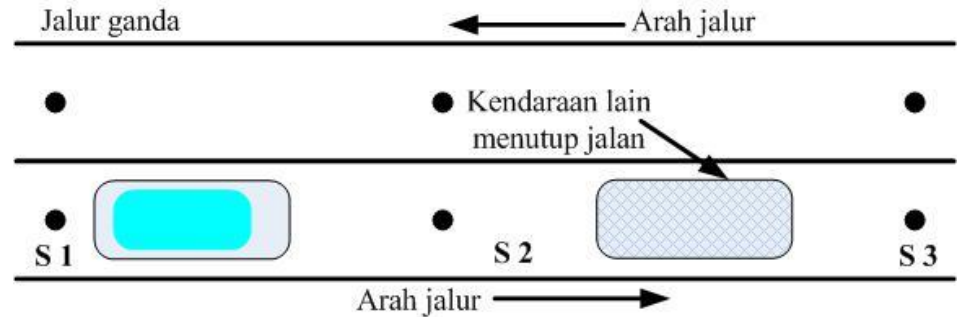
Jalur tunggal



Implementasi HFSM : Skenario 2



Implementasi HFSM : Skenario 3



Implementasi HFSM : Skenario 4

